

HESTIA: A Home Environment Simulator Targeting Inhabitant Activities

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Abstract

The diverse range of Internet of Things (IoT) devices available today offers great potential for home automation and Smart Home (SH) development, whether partial or complete. However, for research in this area to advance, it is essential to overcome the scarcity of data, as data collection in SHs faces significant challenges, such as the complexity of devices and environments, privacy and security concerns, and restrictions imposed by proprietary devices. Furthermore, freely available datasets often have limitations, such as incomplete annotations, restricted device usage, and insufficient spatial and temporal coverage. To address this limitation, we developed a novel SH simulator that supports multi-state devices and captures detailed information on device usage and inhabitant activities, including metrics like lamp brightness and air conditioning settings (temperature, mode, etc.). The primary goal is to provide researchers in SHs, the SH, and related fields with a tool for generating datasets that effectively represent an inhabitant or group of inhabitant activity patterns in simulated environments. Home Environment Simulator Targeting Inhabitant Activities (HESTIA) resolves the inconsistency in message exchanges between devices, which often stems from differences in device manufacturers, while also eliminating the time-consuming and costly process of collecting real-world data by generating synthetic datasets. The simulator HESTIA offers rich, dynamic information that can be tailored to specific user profiles. To validate our method, we collected data over two weeks from a real environment using four devices and compared it to a synthetic version of the same setup generated by HESTIA. The results show that the simulator accurately replicates real-world environments, even for devices with inconsistent status changes. Additionally, we evaluated the computational complexity of this simulator in terms of time.

Keywords

activity simulation, components, circuits, devices and systems, data generator, internet of things, smart environment, smart home

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Abstract—The diverse range of Internet of Things (IoT) devices available today offers great potential for home automation and Smart Home (SH) development, whether partial or complete. However, for research in this area to advance, it is essential to overcome the scarcity of data, as data collection in SHs faces significant challenges, such as the complexity of devices and environments, privacy and security concerns, and restrictions imposed by proprietary devices. Furthermore, freely available datasets often have limitations, such as incomplete annotations, restricted device usage, and insufficient spatial and temporal coverage. To address this limitation, we developed a novel SH simulator that supports multi-state devices and captures detailed information on device usage and inhabitant activities, including metrics like lamp brightness and air conditioning settings (temperature, mode, etc.). The primary goal is to provide researchers in SHs, the SH, and related fields with a tool for generating datasets that effectively represent an inhabitant or group of inhabitant activity patterns in simulated environments. Home Environment Simulator Targeting Inhabitant Activities (HESTIA) resolves the inconsistency in message exchanges between devices, which often stems from differences in device manufacturers, while also eliminating the time-consuming and costly process of collecting real-world data by generating synthetic datasets. The simulator HESTIA offers rich, dynamic information that can be tailored to specific user profiles. To validate our method, we collected data over two weeks from a real environment using four devices and compared it to a synthetic version of the same setup generated by HESTIA. The results show that the simulator accurately replicates real-world environments, even for devices with inconsistent status changes. Additionally, we evaluated the computational complexity of this simulator in terms of time.

Index Terms—Data generator, smart home, smart environment, activity simulation, internet of things

I. INTRODUCTION

By managing and optimizing our daily routines, Smart Homes (SHs) are becoming increasingly relevant in modern life. SHs rely on applications that process in real-time heterogeneous data, often gathered from Internet of Things (IoT) devices. This data enables applications to perceive and understand the surrounding context and interactions, providing valuable insights and even facilitating automated decision-making for these environments [1]. The data collected plays a crucial role in shaping the development of new applications and solutions in this field.

Gathering data from SHs to design new applications and solutions poses significant challenges. First, the inherent com-

plexity and dynamic nature of different environments, along with diverse interactions and ambient variables, complicate the process. Second, privacy and security are real concerns today, and SH data is intrinsically linked to these issues, making it challenging to obtain or share them freely [2], [3]. Third, IoT devices employed to monitor SHs are often proprietary, preventing transparency in monitoring environmental variables and exchanging information. Fourth, available datasets created by the research community are often specific in purpose, lacking heterogeneous data types or offering limited spatial and temporal coverage [4], [5]. This data is fundamental to developing algorithms and models for SHs, and alternatives must be provided to support the development of the next generation of SHs.

Data used to develop SH applications can be either real or synthetic. Real SH data are collected from environments using physical sensors and actuators. However, in addition to the data collection process is challenging due to privacy issues, technical limitations, and financial constraints, real datasets made available for free are limited by factors such as the reduced number of records, incomplete activity annotations, failures, restricted device usage, and data noise. To overcome these limitations, researchers have developed synthetic models designed to simulate the behavior of real sensors in data generation [4]–[9].

Synthetic models allow for the production of datasets that vary across multiple dimensions, including temporal, spatial, and data type diversity. Nevertheless, current synthetic models predominantly simulate basic device functions (e.g., on/off states or numeric readings), resulting in oversimplified scenarios that do not accurately reflect the complexity of real-world environments. For instance, the well-known SH data generator tools like OpenSHS [9] and CASAS¹ are limited to binary and numeric sensor outputs, lack support for actuator devices, and do not yet accommodate scenarios involving multiple human interactions within the environment.

This paper introduces Home Environment Simulator Targeting Inhabitant Activities (HESTIA), an open-source synthetic data generator designed to simulate SHs data. HESTIA generates device messages compliant with formats used by the Tuya² platform, a global provider of IoT devices. This alignment ensures that HESTIA adheres to widely adopted industry

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¹CASAS Synthetic Data Generator: <https://casas.wsu.edu/datasets/sdg.zip>

²Tuya message types: <https://developer.tuya.com/en/docs/iot/message-type?id=Kavqerli65a1u#protocol4>

standards, enhancing its relevance for real-world applications. Additionally, the simulator can generate composite messages with multiple attributes per device, ensuring contextual coherence of human movement and interactions with devices across different rooms. Furthermore, HESTIA supports the creation of multi-inhabitant environments, enabling the simulation of individual routines, activities, and preferences to replicate the behavior of residents in near-real scenarios. In summary, the key contributions of this paper include:

- HESTIA, a validated open-source³ synthetic data generator ready-to-go designed to simulate real-world device data;
- A synthetic dataset validated with real data collected from a smart home;
- A 14-day real-world dataset of a smart environment with multiple devices and data types;
- The environment configuration file used for generating the synthetic data that was validated, allowing the HESTIA user to simulate a larger number of days with the already established and validated routines;
- Time complexity analysis of HESTIA.

The remainder of this paper is organized as follows: Section II defines SH and addresses the complexity of deploying and simulating SH. Section III presents some works related to the research and compares them with our proposal. Section IV explains the structure and the workflow of the HESTIA, how the modeling of devices, environmental spaces, and users works, and how data is generated through the simulation of routines. Section V presents the simulator validation, through visual comparisons using graphs and the Mann-Whitney U hypothesis test, where data generated by HESTIA is compared with data collected from real environments using similar routines. Subsequently, the computational complexity is analyzed in terms of time and memory, and all these details, along with the limitations of HESTIA, are discussed. Finally, Section VI summarizes the work of this paper and presents its problems and challenges.

II. THE COMPLEXITY OF SMART HOMES

Cook et al. [10] define a Smart Environments (SEs) as applications that can acquire and apply knowledge about an environment and also adapt to its inhabitants to improve their experience in that environment. There is a wide variety of SEs, with Smart Homes (SHs) being one of the most researched topics. According to [11], the concept of SH involves the integration of services within a house through a common communication system, ensuring economic, safe, and comfortable operation. Additionally, [12] expands this definition describing the SH as an environment where interconnected devices help inhabitants live independently and comfortably, creating interactive spaces.

However, developing this type of environment is not a simple task. According to [7], [9], [13]–[22], the financial cost is a major challenge in the development and implementation of SHs, as it requires substantial investment in

infrastructure, including sensors, actuators, and the physical environment. Moreover, the authors cite that research-related expenses, such as the collection of datasets through prolonged experiments, further increase the financial burden. The process of developing a SH requires planning due to its complexity and high stakes [9], [13], [15], [19], [21]. Appropriate planning involves determining the optimal placement of sensors, selecting suitable devices, and designing the infrastructure to accommodate automation, all of which demand significant time and resources. Additionally, any errors or oversights in the initial design can make financial costs even higher and, in many cases, infeasible to correct after the physical environment is constructed.

As specified by [7], [13], the scarcity of willing participants with the specific characteristics required for studies is also a major problem hindering the collection of data for SHs research. Additionally, [23] addresses ethics and privacy as critical issues in SH research. The collection of data on human activities, especially in domestic environments, raises concerns about the privacy of residents, as the data can be personal and sensitive. There are also ethical issues related to informed consent from participants, particularly in long-term studies with continuous monitoring.

Data is essential for research in SHs, as it helps understand user behavior and device interaction, contributing to the improvement of automation technologies. The research process in real homes often requires prolonged periods of monitoring and data collection, involving experiments with human activities and system adjustments [9], [15], [19], [21]. Moreover, because of all the challenges mentioned above. The use of open source datasets and SH simulators emerge as solutions to overcome these challenges, allowing for controlled testing and validation of solutions. Another benefit of conducting experiments using simulators and open-source datasets is that these experiments do not suffer from technical issues, such as network failures, device malfunctions, and power outages, which, according to [24], are common sources of frustration for smart device users.

The implementation of a real SH and the simulation require a high degree of flexibility to handle the inherent complexity of such environments [25], [26]. The coexistence of multiple users and devices generates diverse, often conflicting, preferences, needs, and behaviors, while daily routines are unpredictable and frequently involve interleaved or simultaneous tasks. These factors make the development of adaptable solutions essential, enabling the accommodation of variations and the management of overlapping actions, which can obscure individual behaviors and add noise to sensor data.

III. RELATED WORKS

This section summarizes recent literature on available datasets and simulators for Smart Environments (SEs), highlighting their respective simulation approach, which refers to how data is generated. Additionally, two tables detail the main attributes of these studies and compare them with our work. All these attributes are covered by the Home Environment Simulator Targeting Inhabitant Activities (HESTIA) simulator

³HESTIA is available on: <https://github.com/hestia-sim/HESTIA-Smart-Home-Simulator>

and used, throughout this section, to classify other simulators present in the state-of-the-art, but not all of them can be used in the classification of the listed datasets.

- **Simulation Approach (SA)**: is the methodology used for the simulation of SE. Among the works discussed, the SA can be interactive when the user of the generator can interact with the environment; statistical, when the data is generated through statistical distributions or model-based, where the inhabitants are modeled to interact with the environment.
- **Cross-Platform (CP)**: indicates the simulator's availability in different operation systems.
- **Message Type (MT)**: indicates the type of data messages, coming from real or simulator devices. We categorize "composite" based on the versatility of the messages generated by some simulators.
- **Statistical Validation with Real Data (SVRD)**: the simulator was validated with real data through statistical tests.
- **Multi-Inhabitants Support (MIS)**: allows for the modeling of an environment that contains multiple inhabitants.
- **Environmental Modeling (EM)**: the simulator allows you to model different environments, with different numbers of rooms and devices.
- **Activity of Daily Living Based (ADLB)**: the resulting data is performed based on Activity of Daily Livings (ADLs), which, according to [27], are essential self-care activities that support health maintenance and independent living.
- **User Preference Customization (UPC)**: user preferences are modeled and printed on the data generated by the simulator.
- **Open-Source (OS)**: indicates the simulator may be accessed and copied by anyone.
- **Activity Annotation (AA)**: identifies the activities that were being carried out at each moment.
- **Interaction Actor Identification (IAI)**: identifies the inhabitant responsible for interacting with the devices.
- **Inhabitants Group Identification (IGI)**: identifies the group of inhabitants present in the house for each event. The number of identified groups is at most 2^n , where n is the number of inhabitants in the house.

A. Datasets

Table I presents analyzed datasets and compares them based on the attributes listed above. In the following, we present an overview of each related dataset work for SEs. The Activ-

consists of data collected from two environments over thirty days. To build the dataset, the authors integrate devices into the environments without disrupting the users' daily routines, capturing predominantly natural activities. Although this dataset presents two variants, the single user factor and room diversity are limitations. In contrast, HESTIA enables the creation of tailored activity series that align with the specific requirements of the environment, and it is programmable. Furthermore, while the dataset is limited to sensor data, the proposed dataset in HESTIA accommodates the inclusion of actuators, enhancing its applicability for simulating complex interactions in SEs.

The Center for Advanced Studies in Adaptive Systems (CASAS) [28] provides a collection of datasets derived from sensors deployed in different environments. CASAS encompasses 66 datasets from apartments and houses with one up to four residents. Thus, the data is gathered from individuals in their own homes; the authors claim that individuals' daily behavior is captured without environmental or routine activity modifications, resulting in an ecologically valid analysis. The study examines the inter-arrival times of automatically labeled sensor data (e.g., cooking or eating) to analyze activity inter-dependencies.

In [30], a dataset with ADLs recognition using binary sensors. This dataset encompasses data collected from two distinct environments, each occupied by a single user over thirty days. It provides comprehensive information on the timing, operation of sensors, and the specific locations of sensor interactions. From this dataset, it is possible to extract users' routine activities. One significant limitation of the ADLs dataset is the ethical concerns arising from the distribution of devices within the environment. Moreover, the dataset has solely binary data type messages from the devices, which limits the capabilities of applications to derive contextual information and interactions in the SE.

In [31] is provided a dataset comprising monthly registers collected from four distinct countries: Korea, the United States, Spain, and France. The data is structured with dictionaries that provide precise information between the code and values, consisting of time, devices, and controllers from these devices. The available data do not represent user diary routines and cannot indicate the presence of one or more users in the environment. Moreover, the information provided aggregates data from various environments. For example, a routine does not pertain to a specific home or family, making it challenging to discern patterns in routines to predict new device actions.

Meanwhile, our dataset, collected during this study to validate HESTIA, presents a residential room inhabited by a single person over two weeks, from August 3 to August 19, 2024. The bedroom, being a multifunctional space, serves both work and leisure purposes, with a computer as the focal point of the monitored interactions. Data collection spanned weekdays and weekends and was conducted through actuators such as a lamp, the computer's plug, and a fan, as well as sensors that recorded presence, window opening, and environmental conditions like temperature and humidity. Each of these devices generated messages containing one or more attributes in JSON format. Notably, the presence sensor exhibited some noise, occasionally misreporting the inhabitant's absence when

TABLE I
MAIN CHARACTERISTICS EXTRACTED FROM RELEVANT
STATE-OF-THE-ART DATASETS.

Tool/Author	MT	MIS	ADLB	OS	AA
CASAS Dataset [28]	Binary/Numeric	Yes	Yes	Yes	Yes
ARAS [29]	Binary	Yes	Yes	Yes	Yes
Chawathe:2018 [30]	Binary	Undefined	Yes	Yes	Yes
SmartThings [31]	Composite	Undefined	No	Yes	No
Our real dataset	Composite	No	Yes	Yes	No

ity Recognition with Ambient Sensing (ARAS) dataset [29]

they were present, and vice versa. On weekdays, the inhabitant followed a routine, leaving for work and returning in the evening, whereas weekends allowed for greater flexibility in activities, with a less predictable presence pattern. A total of 33,338 messages were recorded, reflecting the interactions between the inhabitant and the devices during this period.

B. Simulators

In addition to datasets, several tools have been developed to generate data for SEs, often complementing the use of real-world data. Table II presents a comparative overview of these simulators. We then provide an overview of each related work on simulators for SEs.

In [20], the authors introduce PerSim3D, a data generator tool designed for smart environments (SEs) that represents activities as sequences of actions performed by a virtual character interacting with various objects, potentially altering their states. The tool is a successor of [13] and has a Graphical User Interface (GUI) for 3D activity visualization and generates sensor data using the Sensory Dataset Description Language. However, as drawbacks, PerSim3D is not publicly available, limiting its accessibility for the community of developers of SEs application and researchers, the simulator only produces binary data types, it allows only single inhabitants per environment, and it is not possible to programming User Preference Customization and do annotations to the data produced.

OpenSHS [9] is an open-source application developed to simulate smart homes. It can produce synthetic data for various sensors, such as pressure, door, and light switches. The tool uses Blender and Python scripts, allowing users to design custom three-dimensional environments for modeling daily routines. Although OpenSHS has several strengths, it presents some limitations when compared to HESTIA. It only supports binary (on/off) device statuses and single inhabitant simulation, restricting its ability to simulate complex environments with multiple occupants, sensor interactions, and different user preferences. These limitations pose challenges when trying to model more sophisticated smart home dynamics.

Francillette et. al. [7] underscores the limitations of existing datasets in adequately capturing user behavior in a SE. The authors argue that while synthetic data generated by simulators does not need to perfectly mirror real-world data, it must encapsulate essential behavioral patterns and concepts. Their methodology employs tree-based models to represent user behavior, offering a detailed setup for environmental and sensor configurations. Additionally, they present an open-source tool that uses Unity3D, which simulates smart home environments with support for multiple inhabitants and daily routines, to produce synthetic SE data. However, this tool is deprecated, making it hard or impossible to install it due to system library compatibility issues. Also, the tools do not allow User Preference Customization and data annotations. In contrast, HESTIA takes a graph-based approach to model user interactions with devices within a space, focusing specifically on routine behaviors and offering a flexible framework for simulating user-device interactions across various environmental setups. Also, HESTIA is implemented in Python, making it a multiplatform tool and ready to go.

In [6], the authors proposed SEsim. The simulator can simulate four distinct areas, generating synthetic data from various artificial sensors such as proximity, light, temperature, and touch. The simulator can generate single individual daily routines. Also, the authors state that SEsim has a GUI implemented in Unity3D. As drawbacks, SEsim does not support multi-inhabitants, ADLs, and UPC. Additionally, SEsim is not open-source, which limits its accessibility to the community. SEsim shares similarities with HESTIA in generating sensor data and recognizing user activities. But HESTIA focuses on modeling daily activity distributions to predict device behavior consistently across routine patterns.

ESEsim [32] is an extension of SEsim. Building on the original system, ESEsim expands its capabilities by supporting environments shared among multiple occupants. ESEsim aims to generate synthetic data that can be employed for Human Activity Recognition (HAR), providing data suitable for machine learning models. The authors also present machine-learning techniques for recognizing activities such as personal hygiene, dressing, cooking, and sleeping. Like SEsim, ESEsim incorporates a stochastic simulation model, maintaining random activity variability while expanding support for multiple users. As drawbacks compared to the HESTIA simulator, ESEsim is not based on ADLs and does not support UPC. Additionally, ESEsim is not an open-source tool.

Synthetic data generation models face the challenge of balancing implementation simplicity with the capacity to accurately replicate real-world environment data. SynSys [33] addresses the trade-off by employing Hidden Markov Models (HMMs) and regression models, which are trained on real-world datasets to produce synthetic time series data. The authors validated the method using an annotated smart home dataset. The authors apply Dynamic Time Warping (DTW) to measure the similarity between real and synthetic data. Each activity in SynSys is represented by its own HMM, generating sequences of sensor events associated with that activity. A second-level HMM refines these events into detailed sensor sequences, while regression models assign precise timestamps and activity durations. Conversely, HESTIA focuses on device-level predictions and provides a flexible abstraction for modeling environments, while SynSys centers on activity-driven data generation.

In [5], the authors propose Virtual Smart Home (VSH) tool, a model-based simulator for SEs. It produces synthetic data on the daily activities of a smart home. VSH was built upon the Virtual Home simulator⁴ integrating the logging of sensor activations in a smart home environment. However, unlike HESTIA, this simulator only works with binary data, does not support multiple inhabitants, and consequently does not account for user preferences. To evaluate the realism of the simulation, the authors compared the simulated activities with a multi-inhabitant dataset collected from a real living lab. The validation process focused on comparing the frequency of sensor activations between the synthetic scenario and the real-world data, providing insights into the tool's ability to accurately replicate smart home interactions.

⁴Virtual Home is available on: <http://virtual-home.org>

TABLE II
MAIN CHARACTERISTICS EXTRACTED FROM RELEVANT STATE-OF-THE-ART SE SIMULATORS.

Tool/Author	SA	CP	SVRD	MT	MIS	EM	ADLB	UPC	OS	AA	IAI	IGI
Persim3D [20]	Model-Based	Yes	No	Binary	No	Yes	Yes	No	No	Undefined	No	No
OpenSHS [9]	Interactive	Yes	No	Binary	Yes	Yes	Yes	No	Yes	Yes	No	No
Francillette:2017 [7]	Model-Based	Yes	No	Binary/Numeric	Yes	Yes	Yes	No	Yes	No	No	No
SESim [6]	Model-Based	Yes	No	Undefined	No	Yes	No	No	No	Undefined	No	No
ESESim [32]	Model-Based	Yes	No	Undefined	Yes	Yes	No	No	No	Undefined	No	No
SynSys [33]	Statistics	Yes	Yes	Binary	Yes	No	Yes	No	Yes	Yes	No	No
Virtual Smart Home [5]	Model-Based	Yes	Yes	Binary	No	Yes	Yes	No	Yes	Yes	No	No
Veneruso:2023 [8]	Model-Based	No	No	Binary/Numeric	Yes	Yes	Yes	No	No	Yes	No	No
HESTIA	Model-Based	Yes	Yes	Binary/Composite	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

In [8], the authors propose a simulator designed to generate SEs synthetic data. The simulator offers a variety of scenarios, allowing researchers to create benchmarks tailored to different sensors and environmental contexts. Moreover, it simulates human activities within smart homes. A key feature is the collaborative component, which includes a public repository where the community can contribute with new models, boosting the creation of synthetic open-access datasets. In comparison to HESTIA, a notable distinction is that, in our approach, one can perform User Preference Customization, and HESTIA is an open-source tool. Additionally, while the Veneruso et al. [8] simulator uses an extensible event stream format for output, this format is less commonly adopted in scientific datasets that focus on pattern recognition, whereas HESTIA follows industry output format standard.

To the best of our knowledge, HESTIA is the pioneering tool capable of managing both simple and composite messages, enabling the simulation of diverse environments and devices. It supports a wide range of devices, from basic ones (e.g., light switches) to advanced equipment (e.g., smart TVs and air conditioners), extending simple on/off status changes to multi-level status (e.g., light dimmer and color). HESTIA also allows program usage preferences, enabling users to customize tasks or devices according to specific needs. It supports multi-inhabitant scenarios, providing a more realistic simulation of real-world conditions. Furthermore, activity-based modeling can incorporate interconnected and independent activities, allowing task scheduling based on predefined statistical probabilities and related actions. Finally, HESTIA is an open-source and ready-to-go multiplatform tool.

IV. HESTIA ARCHITECTURE AND WORKFLOW

This section describes the architecture of Home Environment Simulator Targeting Inhabitant Activities (HESTIA), its components, and its workflow as shown in Figure 1. HESTIA is composed of two main parts: the *back-end* and the *front-end*. The former, the focus of this paper, is responsible for manipulating, processing, and exporting synthetic data. The second part, *front-end*, allows users to interact with HESTIA through a Command Line Interface (CLI).

In Figure 1, each component presents the fundamental activities performed by HESTIA. The framework is implemented in Python, and its architecture allows easy extensions, enhancements, and maintenance. The HESTIA has the MIT license, and it is available in an online repository³. Each component of the architecture is detailed below.

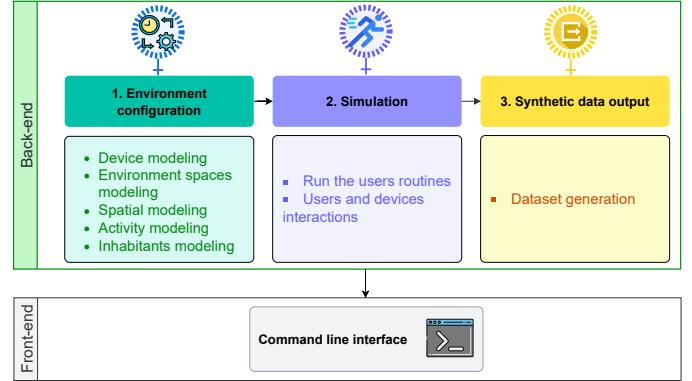


Fig. 1. HESTIA architecture and workflow.

In a nutshell, the HESTIA's workflow is described as follows: 1) **The Smart Environment (SE) configuration phase**: One can configure the environment by setting the number, interconnections, and characteristics of rooms. The smart devices, including sensors and actuators, can be customized in terms of number, types, behavior (data generated, types, frequency, etc.), and placement. Finally, the number of simulated inhabitants and their routines can be set; 2) **Simulation phase**: the pre-programmed inhabitants routines and sensor behaviours of simulated SE are executed at this step. As a result, synthetic sensor data can be produced over time and by simulated inhabitants' interactions with sensor and actuators; 3) **Synthetic data output phase**: the produced data is formatted into CSV files, where each register refers to an inhabitant's interaction with the home. The latter is built to be industry compliant, fitting to the standard data scheme for message exchanges by real sensors and actuators smart devices.

A. Device modeling

In HESTIA, devices are modeled and organized within a hierarchical structure, in which each device is a concrete instance of a *Product* class, which, in turn, represents a specific model, type, or group of device. *Product* is a class that defines minimum abstract functions, attributes, and method behaviors for each type of device. For example, we may have multiple lamp devices spread throughout the house: bedroom light bulb, living room light bulb, bathroom light bulb, and so forth; where each of them is an instance of the lamp *Product*. The HESTIA approach enables flexible and extensible device

modeling, accommodating the diverse needs of HESTIA users. We provide HESTIA with 7 implemented Products: coffee machine, light bulb, presence sensor, smart plug, sound system, air-conditioner, and TV, allowing their real-world interactions and behaviors to be simulated. Contributors are welcome and free to add new Products depending on their needs.

The attributes of a Product class represent devices' operational conditions or capacities. For instance, device status such as "switch", "temperature" and "mode", Figure 2a. Its values depend on the features that a product has. Also, devices are programmed to send devices' status when an interaction happens, in a standard format. Figure 2b presents an example of an air conditioning device' status. Each status message has at least a code referring to a device feature (code), a timestamp labeling when an of the occurrence (t), and the current feature value (value).

This approach enables accurate simulation by allowing device programmability and real device attribute reproduction. Also, it is possible to extend HESTIA to accommodate new products as users need. The extensibility of the HESTIA enables the creation of new product classes before the simulation begins, ensuring that the platform adapts to technological advancements and diverse smart environment configurations and necessities.

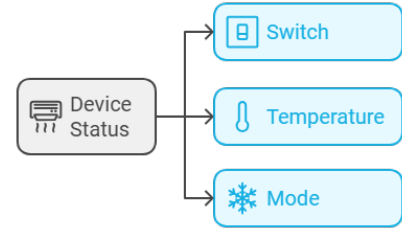
B. Environment spaces modeling

To effectively model the environment, HESTIA relies on several key components. One such component is shown in Figure 3, which illustrates a fundamental building block for this process. The environment encompasses a list of spaces, an abstraction of rooms within an environment. Each space has a name (e.g., kitchen, office, bedroom, bathroom, etc), a list of associated sensor/actuator devices, and modeling parameters. In the simulation environment configuration, spaces are created with a device list allocated to them, where each item in this list is a Product (seen in Subsection IV-A), which is an abstraction of a device in HESTIA simulation. Additionally, each space has a maximum occupancy parameter, which controls access to each space. Thus, it is possible to restrict the maximum number of inhabitants in a space when necessary. For example, the bathroom may be restricted to one occupant at a time to reflect realistic usage, while the kitchen can accommodate up to three individuals simultaneously, representing typical scenarios such as meal preparation or group cooking activities.

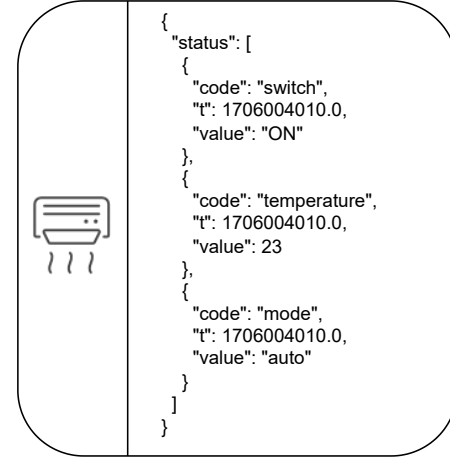
The number of spaces in each simulation corresponds to the number of rooms in the target environment, plus one, which represents the street (an external space with no associated devices). This external space is included to account for moments when inhabitants are outside the home.

C. Spatial modeling

Spatial modeling of rooms ensures an accurate correlation of data with the positioning of inhabitants interacting with devices in the smart environment. To capture these movements, a weighted graph structure is employed to model the distribution of spaces within the simulated home. In this model, each node



(a) Example of programmed device's status.



(b) Structure of JSON messages for a smart devices.

Fig. 2. In HESTIA, (a) one can program the device behavior; and (b) timed or by event message status reports.

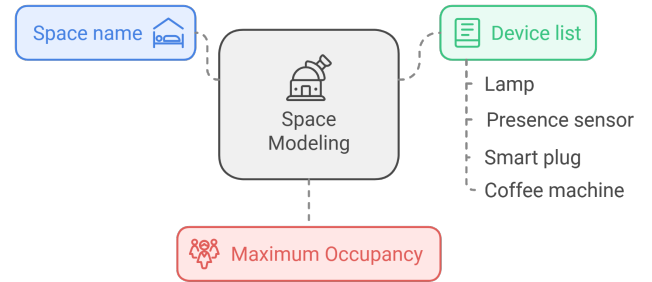


Fig. 3. HESTIA spaces modeling approach.

represents a room, such as a bedroom, bathroom, and kitchen. The edges represent the possible paths between the spaces, according to the physical configuration of the environment. The weights represent the travel time, in seconds, between two spaces (nodes).

Figure 4 illustrates how the spaces of a simulated smart home are modeled in HESTIA simulation. Each connection is created through three-element lists that specify the two connected spaces and the weight between them. In this graph, the A^* algorithm is used for pathfinding. This algorithm com-

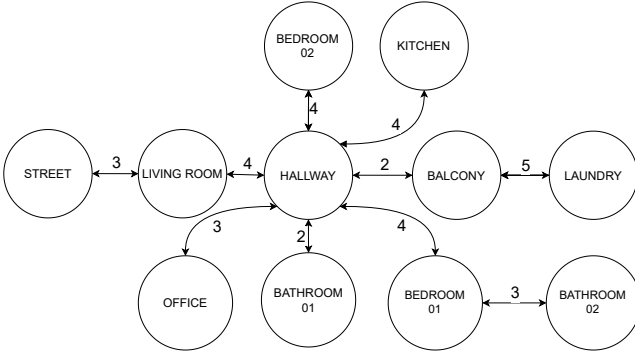


Fig. 4. Graph of spaces distributed in a smart environment weighted as time in seconds for room displacement.

bin heuristic approximations with the formality of Dijkstra's algorithm and is one of the most used in this type of task.

This modeling allows HESTIA users to adjust the simulation behavior according to the spaces' occupancy capacity, the detection range of the space presence sensors, and the distribution of spaces in the house to be simulated. This allows the spatial structure to be customized to simulate smart homes with different configurations. Additionally, the graph representation enables the representation of the movement of inhabitants within the house, taking into account both the proximity between spaces and the restrictions imposed by the physical structure, which is essential for generating data representative of real interaction scenarios in a smart home.

D. Activity modeling

The stage of activities modeling is crucial in HESTIA, as it is performed based on Activity of Daily Livings (ADLs), and the entire simulation is based on the activities performed by the inhabitants. In this stage, the HESTIA's user is free to define as many activities as necessary. Additionally, HESTIA allows the association of activities, enabling shorter-duration activities to take place within the execution time of longer-duration activities. Therefore, a more detailed simulated routine requires specifying a greater number of activities.

In the activity creation process, a sequence of fields must be defined to ensure its proper execution, as can be seen in Figure 5. The *name* field identifies the name of the activity. The *duration* field represents the duration of the activity in minutes, while the *time_variation* key adds variability in the execution time, being represented by a percentage. For example, in an activity named *USE_PC_NIGHT*, with *duration* equal to 60 minutes and the *time_variation* value equal 5%, the real duration for this activity may be any value between 57 and 63, simulating realistic inaccuracies.

In addition, the *associated_activities* field allows the definition of other activities that can be performed during the activity in processing. The associated activities share the same hierarchical level as the activities being processed, and their duration must be shorter than that of the latter. Moreover, each associated activity has a probability of occurrence value, where the higher the value, the greater the probability of the

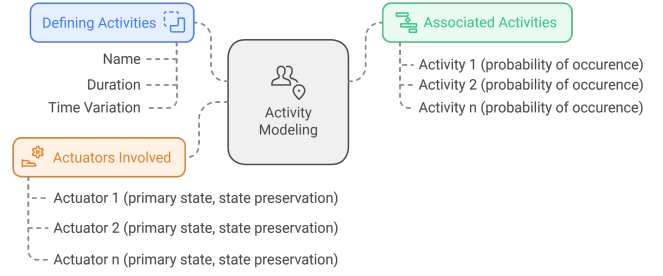


Fig. 5. HESTIA activity modeling fields.

associated activity occurring during the running activity. For example, if the activity *USE_PC_NIGHT* has an association activity *USE_BATHROOM* at a 2% probability of occurrence, this means that while performing *USE_PC_NIGHT*, the inhabitant has a 2% chance of switching to the activity *USE_BATHROOM* and after returning to *USE_PC_NIGHT*. But, *associated_activities* is not a mandatory field, if the HESTIA's user does not want to associate any activity with the activity he is modeling, this field can be neglected.

Finally, the *list_actuators_activity* field specifies all the actuators involved in the execution of the activity, defining the *primary state* of each device during the activity and the *state preservation*, indicating whether the device's state will be preserved in the event of an associated activity. When defining the *primary state* of the device, the user needs to specify whether it will be turned on or off during the execution of the activity. By defining *state preservation*, the user must specify whether, while the inhabitant performs one of the associated activities, the device will assume a deactivated state (represented by "D"), turning "OFF", or if it will preserve its current state (represented by "P"), maintaining the states of the devices used by the primary activity.

This modeling allows simulating the variability of the activities performed by residents, reflecting common behaviors, such as pausing one activity to perform another. This type of variation is common in everyday life and is usually caused by external factors, contributing to a simulation that is closer to reality and more dynamic.

E. Inhabitants modeling

Modeling inhabitants is a fundamental step in HESTIA, once the simulation process is based on the inhabitant's behavior and interactions within the simulated smart home. This simulator supports multiple inhabitants, where each inhabitant is represented individually with 5 attributes: *name*, *initial space*, *preferences*, *priority* and *weekly routine*. The *name* field is used to identify the user in the simulation. The *initial space* field specifies the location within the environment where the user starts at the beginning of the simulation. The *preferences* field defines the inhabitant preference profile through their choices regarding specific devices, allowing users to adjust the settings according to their needs. For instance, one user might prefer the lighting in their bedroom to be adjusted to a softer, warmer shade of light at night, while another user might prefer a brighter, lower light at dawn. Additionally, the

air conditioning can be set to maintain a specific temperature when the user is present, and the TV can automatically adjust the volume and channel selection based on defined preference profiles.

Sometimes there may be competition for the use of devices by different inhabitants present in the same space. Thus, *Priority* field is employed to determine the precedence of device use. For instance, if inhabitant A has a higher priority than inhabitant B, then in cases where both compete for the use of the same device, inhabitant A's preferences will be selected by the simulator instead of B's preferences. The *week routines* is represented by a jagged array, as it provides flexibility to handle irregular data, where each day can have a different number of activities. The structure consists of a 7-line array, where each line corresponds to a specific day of the week, from Monday to Sunday, and specifies the sequence of activities performed during the simulation by the inhabitant. The activities are organized sequentially, with the first item marking the start of the routine at midnight (00:00) and subsequent activities positioned based on their duration. It is important to highlight that HESTIA handles activities that span from one day to another by splitting the activity duration suitably for the days.

While modeling routines are extensive, it enable detailed control over the activities performed during the simulation, allowing the user to tailor the inhabitant's routines to specific needs. It is important to note that the definition of these routines has a significant impact on the data generated and on the system's ability to produce realistic results. If a routine is "poor", with few activities or low granularity, the resulting data will have limited variation and interactions. Therefore, it is up to the user to create routines that accurately represent the environment's use, ensuring the quality of the data produced.

F. Simulation

In the simulation process, HESTIA uses the SimPy library⁵, a process-based discrete-event simulation framework written in standard Python. SimPy supports the modeling of active components such as clients, vehicles, and agents. In HESTIA, we use this library to discretize time on a second-by-second basis, managing events and controlling resources, including spaces and devices. Events are generated based on each user's weekly activities, prioritizing their preferences. Each user is processed in parallel, allowing their weekly activities to be executed as individual sequences. During this phase, conflicts may arise when multiple users compete for the same device or space. Conflict resolution is carried out based on the priority assigned to each user.

To perform the simulation, We developed a logic that accounts not only for the activities performed by simulated inhabitants but also for their preferences, the presence of multiple simulated housemates, and the need for dynamic device adjustments based on interactions within the space. The flowchart shown in Figure 6 illustrates the logic implemented in HESTIA, detailing how inhabitant activities and preferences influence the behavior of devices in the environment.

To run a simulation, one may provide a custom-made configuration that informs HESTIA about what to simulate. This configuration includes essential details such: 1) the number of smart devices, inhabitants, and rooms in the environment (see Sections IV-C and IV-A); 2) the behavior of smart devices and simulated inhabitants (see Sections IV-E and IV-D). The user also specifies the duration of the simulation by indicating the number of simulated days.

The simulation sequence begins on the first day by selecting the next (or first) activity of a simulated inhabitant. This is followed by selecting the action to be taken and the location where it will occur. After defining the activity, the system checks if the inhabitant is in the space where the activity needs to be performed. The process advances to the next step if the inhabitant is already in the location. Otherwise, the system uses the environment graph to calculate the route to the activity location and simulates the movement to that space. During this movement process, the presence sensors along the route are activated as the inhabitant traverses the environment.

The system also checks if there are other inhabitants in the same space. If no other inhabitants are present, the simulator selects the necessary devices for the activity based on the established settings. If there are other inhabitants, the system checks whether they are using the devices required by the current inhabitant. If the devices are in use, the simulator examines whether these inhabitants have configured preferences. If no preferences are set for the other inhabitants, the system then considers the preferences of the current inhabitant. However, if the other inhabitants have preferences, the system compares them with the current inhabitant's preferences to resolve any conflicts, determining which preference should prevail based on a pre-established order of priority. This configuration allows certain inhabitants to have predominant preferences over others, ensuring a hierarchy in device use.

After resolving any inhabitants' preference conflicts, the simulator checks if the current states of the devices meet the activity requirements. If the states are appropriate, the activity proceeds; otherwise, the system adjusts the devices accordingly to perform the activity.

HESTIA can simulate subactivities, reflecting human behavior more accurately. For example, it is common for a person to begin an activity, such as watching TV, then engage in a short subactivity, like going to the bathroom, before returning to the main activity, such as resuming TV watching. To do so, HESTIA verifies if there are any associated activities linked to the inhabitant's current activity. Associated activities can happen randomly every 10 minutes of an activity simulation. If no associated activity is selected, the inhabitant remains in the space until the main activity is completed. Otherwise, e.g., if an associated activity is selected, the resident will perform it returning to the primary activity once finished. At the end of each activity, whether the primary or an associated one, the simulator restores the devices to their previous status before the activity begins.

Once the activity is processed and all associated activities have been completed, the system checks whether the total number of simulated days reaches the pre-configured days of simulation required, redoing the loop as needed.

⁵SimPy is available on: <https://gitlab.com/team-simpy/simpy>

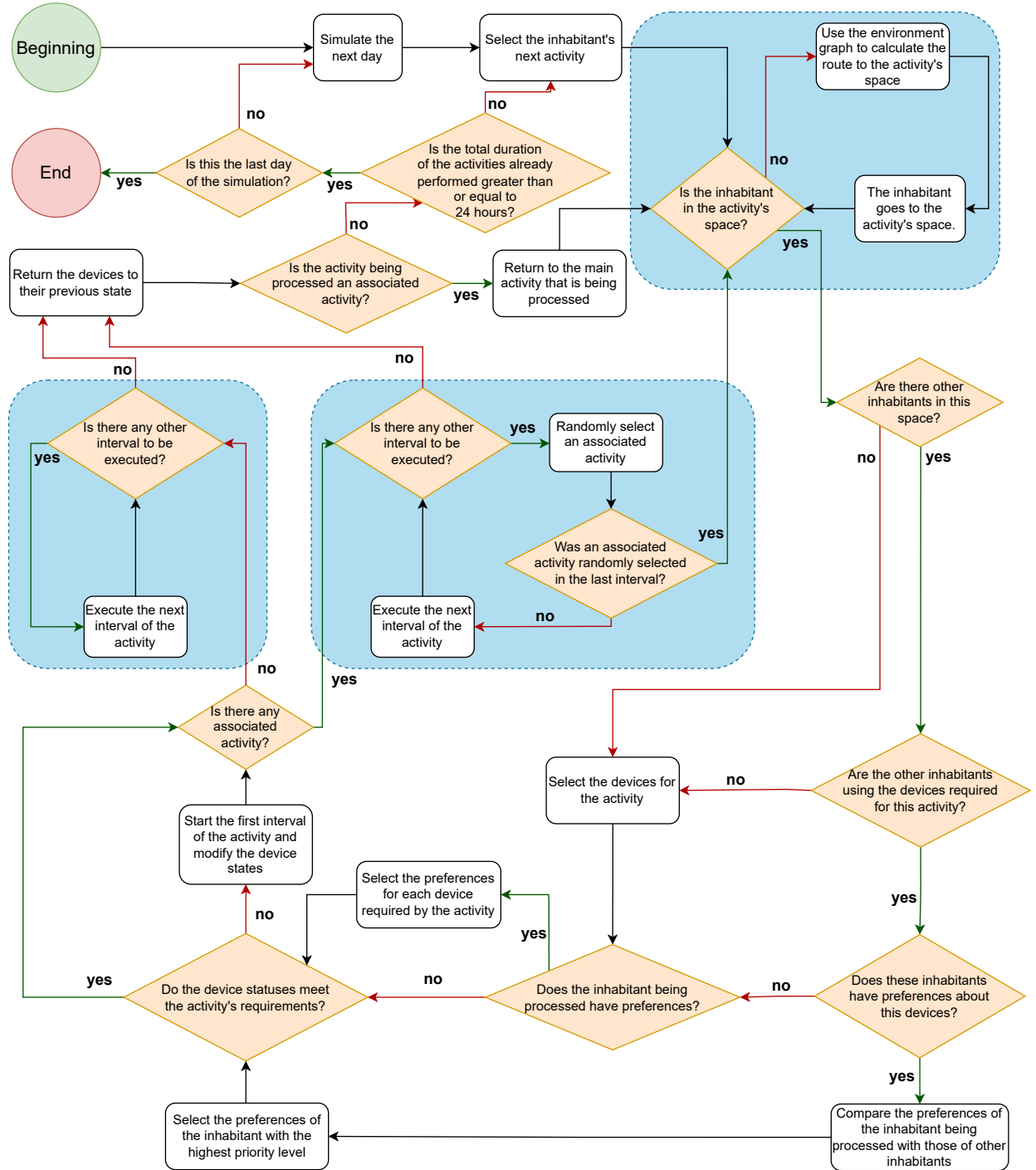


Fig. 6. Flowchart of events generation, considering interactions with users and devices. The blue blocks represent operations executed during simulation time, based on time unit defined in SimPy, such as performing activities or users moving through the environment.

This logic, used in HESTIA, aims to replicate realistic interactions in a smart home environment. It considers factors such as inhabitants' movement, individual preferences, usage conflicts between residents, and dynamic adjustments to devices based on the needs of each activity. Thus, the simulator provides a personalized representation of how multiple inhabitants interact with shared devices in a dynamic, interactive smart environment.

G. Dataset generation

During the simulation of a smart home, data is collected throughout the process. As a result, it is possible to create a dataset that can be used for experimentation and evaluation in research. Table III shows a sample of the generated dataset. It highlights the behaviors and activities resulting from the simulated interactions between the inhabitants and the devices of the smart home. The columns of the resulting dataset are presented below:

TABLE III

DATASET REPRESENTING THE FORMAT OF THE DATA GENERATED IN CSV THROUGHOUT THE HESTIA SIMULATION. THE FORMAT OF THE VALUES IN THE “MESSAGE” COLUMN IS SHOWN IN FIGURE 2B, WHICH REPRESENTS THE VALUE OF “MESSAGE 1”.

TimeStamp	Device	DevId	ProductKey	Message	SensorType	Group	UserAction	ActivityUserAction	Space
2024-01-23 10:00:10.000000	BEDROOM_PRESENCE_SENSOR	d1	p1	message 1	sensor	[Viktor, Jayce]	Jayce	SLEEP	BEDROOM
2024-01-23 20:00:00.271228	BEDROOM_PRESENCE_SENSOR	d1	p1	message 2	sensor	[Viktor, Jayce]	Viktor	USE_COMPUTER	BEDROOM
2024-01-23 20:00:03.321280	BEDROOM_LAMP	d2	p2	message 3	actuator	[Viktor, Jayce]	Viktor	USE_COMPUTER	BEDROOM
2024-01-23 20:00:05.273201	BEDROOM_PLUG	d3	p3	message 4	actuator	[Viktor, Jayce]	Viktor	USE_COMPUTER	BEDROOM
2024-01-24 00:00:46.422832	KITCHEN_PRESENCE_SENSOR	d4	p1	message 5	sensor	[Viktor, Jayce]	Jayce	DRINK_WATER	KITCHEN
2024-01-24 00:00:47.990696	KITCHEN_LAMP	d5	p2	message 6	actuator	[Viktor, Jayce]	Jayce	DRINK_WATER	KITCHEN
2024-01-24 00:03:31.266813	KITCHEN_LAMP	d5	p2	message 7	actuator	[Viktor, Jayce]	Jayce	DRINK_WATER	KITCHEN
2024-01-24 00:03:32.828371	KITCHEN_PRESENCE_SENSOR	d4	p1	message 8	sensor	[Viktor, Jayce]	Jayce	SLEEP	KITCHEN
2024-01-24 00:10:16.145221	BEDROOM_PLUG	d3	p3	message 9	actuator	[Viktor, Jayce]	Viktor	USE_COMPUTER	BEDROOM
2024-01-24 00:12:31.422113	BEDROOM_LAMP	d2	p2	message 10	actuator	[Viktor, Jayce]	Viktor	USE_COMPUTER	BEDROOM
2024-01-24 00:51:24.145848	BEDROOM_AIR_CONDITIONING	d6	p4	message 11	sensor	[Viktor, Jayce]	Jayce	SLEEP	BEDROOM
2024-01-24 00:51:24.145848	BEDROOM_PRESENCE_SENSOR	d1	p1	message 12	sensor	[Viktor, Jayce]	Jayce	SLEEP	BEDROOM
:	:	:	:	:	:	:	:	:	:

- **TimeStamp**: the exact moment when the message was recorded, in date and time format.
- **Device**: the name of the device that captured or performed the action.
- **DevId**: the unique identifier of the device that acted.
- **ProductKey**: the key or identifier of the product to which the device belongs, used to identify the specific type of device.
- **Message**: message associated with the recorded action, providing specific details about the event and device state.
- **DeviceType**: type of device (sensor or actuator), indicating whether the device collects data or performs an action.
- **Group**: a list representing the group of people who were present in the house at the time the event was recorded, indicating those who had access to or could interact with the specified device.
- **ActorInhabitant**: the inhabitant who performed the recorded event.
- **Activity**: the specific activity or action simulated at the recorded time.
- **Space**: the space or room where the action took place.



Fig. 7. Real scenario representation used to capture device information during fourteen days.

TABLE IV
DETAILS OF THE MONITORED DEVICES.

device	example of a message
Plug_fan	[{'code': 'switch_1', 'dpld': 1, 'time': 1722739237311, 'value': True}]
Plug_pc	[{'code': 'switch_1', 'dpld': 1, 'time': 1722739266900, 'value': False}]
Humidity_sensor	[{'code': 'humidity_value', 'dpld': 2, 'time': 1722739282372, 'value': 71}]
Presence_sensor	[{'code': 'presence_state', 'dpld': 1, 'time': 1722739300319, 'value': 'none'}]
Light_bulb	[{'code': 'switch_led', 'dpld': 20, 'time': 1722742849952, 'value': False}]
Window_opening_sensor	[{'code': 'doorcontact_state', 'dpld': 1, 'time': 1722744971500, 'value': False}]

V. EXPERIMENTAL EVALUATION

In the experimental phase, Home Environment Simulator Targeting Inhabitant Activities (HESTIA), the Smart Home (SH) simulator, was rigorously evaluated by comparing its synthetic data outputs with real-world data collected from a smart environment. The validation of HESTIA was carried out in three phases: initially, we collected data from a real environment using actual smart devices. Next, we used HESTIA to configure the same environment and simulate the behavior of the users and devices present. Finally, we compared the data generated in the real environment and in HESTIA, presenting, throughout this section, not only graphical visualizations but also the results of the Mann-Whitney U hypothesis test, conducted at a 95% confidence level, to test the hypothesis that the frequency distributions of the data generated by HESTIA and the real-world data are similar.

A. Real smart home environment scenario

Over 14 days, we collected real-world sensor data from a single resident in a smart room. As shown in Figure 7,

the room included a table, chair, computer monitor, bed, and wardrobe, representing a typical work-and-living environment.

In this environment, we installed various smart devices commonly used in smart home systems. Table IV shows the list of smart devices in this experiment and how their messages are returned. These devices were utilized to collect status data signals throughout the monitoring period.

The devices monitored the resident’s daily routines, recording activities such as presence detection, lighting usage, fan operation, and computer activity. Over the 14 days, a total of 33,338 messages were generated. Among these, 29,690 were report messages, capturing status updates and electrical current fluctuations, while 3,608 messages directly reflected user interactions with the devices.

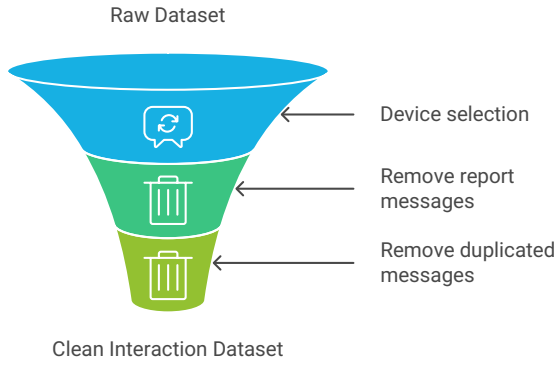


Fig. 8. Data-cleaning process.

TABLE V
DATA DISTRIBUTION BEFORE AND AFTER EACH FILTERING STEP.

Device	Raw	Reports	Interactions	Interactions without duplicates
Sensor_presence	17345	12915	4430	3386
Light_bulb	159	20	139	134
Plug_pc	12196	12146	50	50
Plug_fan	2143	2105	38	38
Total	31843	27186	4657	3608

B. Data cleaning

The data collected from the real environment served as the basis for modeling the synthetic routines generated by the simulator. Given that HESTIA focuses on simulating interactions between residents and devices, cleaning the raw data coming from the real devices was essential to ensure meaningful comparative analysis. In Figure 8, the data-cleaning process for the collected real-world data can be observed.

We began by selecting the devices that would be simulated: the presence sensor, smart lamp, smart plug controlling the fan, and smart plug connected to the personal computer. These devices were selected because the product classes for them are already implemented in HESTIA. Thus, out of a total of 33,338 messages, only the messages generated by these four devices remained, totaling 31,843. Later, we removed the report messages, such as status check messages and electrical current data. Finally, we removed duplicate messages, such as two or more consecutive messages bringing the same information, for instance, indicating more than once that a lamp was turned on. Table V shows the distribution of messages before and after data cleaning.

After the data cleaning process, we obtained 3,608 interaction messages, of which 3,386 were attributed solely to the presence sensor. This highlighted the importance of careful analysis to identify and filter out irrelevant events during the data modeling and validation process.

C. Analysis of the devices' message frequency

Figure 9 shows the average frequency of messages generated per device per hour. This graph compares the average number of messages generated over 14 days in the real dataset with data generated by HESTIA for the same period. In it, you can see interactions such as state changes in the presence sensor, light bulb, and smart plugs.

It is possible to notice a similar trend among the three devices, while the presence sensor shows slightly different behavior. Unlike the other three devices, the presence sensor changes its state even when there is no real change in the physical environment, that is, when the resident remains in the same place, generating numerous false positives. For example, on several occasions during data capture, when the inhabitant was present in the room, the presence sensor indicated his/her absence, and on other occasions, when the inhabitant was absent, the sensor indicated presence. Even when the resident was present, the presence sensor frequently toggled between “presence” and “none” states, creating conflicting data and complicating the filtering of relevant information.

To test the hypothesis arising from the analysis of these results, we performed the statistical Mann-Whitney U test to evaluate the similarity between the real and generated data regarding the number of messages generated by the devices. Where we compare the distribution of device messages between the 14-day generated and real data and the real data for the same period. In it, all the collection days were compared and analyzed, hour by hour, checking the number of generated messages for each device. The Mann-Whitney U test compares two independent groups to determine if there is a significant difference in their distributions.

Table VI shows the results of the Mann-Whitney U test for each device analyzed. The results indicate that there is a statistically significant difference between the generated and actual data only for the presence sensor, where the p-value was less than 0.05. For all other devices, this value was greater. Therefore, we conclude that HESTIA successfully simulated message generation for three different devices.

As mentioned in Section II, one of the biggest challenges when working with real smart devices is the generation of noisy data. However, the difference in the number of messages generated by a real presence sensor and a simulated one is particularly noticeable, as the goal of HESTIA so far has not been to simulate noisy data.

D. Analysis of the devices' active time

To reduce discrepancies caused by device noise in the real environment, we focused on analyzing the period during which each device remains active, as shown in Figure 10. This period was calculated by counting the active hours for each device in each hour interval. We consider the active time of the devices as follows: the time during which the presence sensor detects the presence (*presence_state* = *presence*), and the lamp and plugs are turned on (*switch* = *ON*). For example, if device A was turned on at 8:30 and off at 9:20, we considered the following: in the 8:00 hour interval (referring to the period between 8:00 and 9:00), the device stayed on for 30 minutes, and in the 9:00 hour interval (between 9:00 and 10:00), it stayed on for 20 minutes. That is, we consider 24 one-hour intervals throughout the day. Thus, we calculate the average time each device stays active during each of these intervals. In each of these intervals, a device can register between 0 and 60 minutes in this state, depending on the average time it was active during the period.

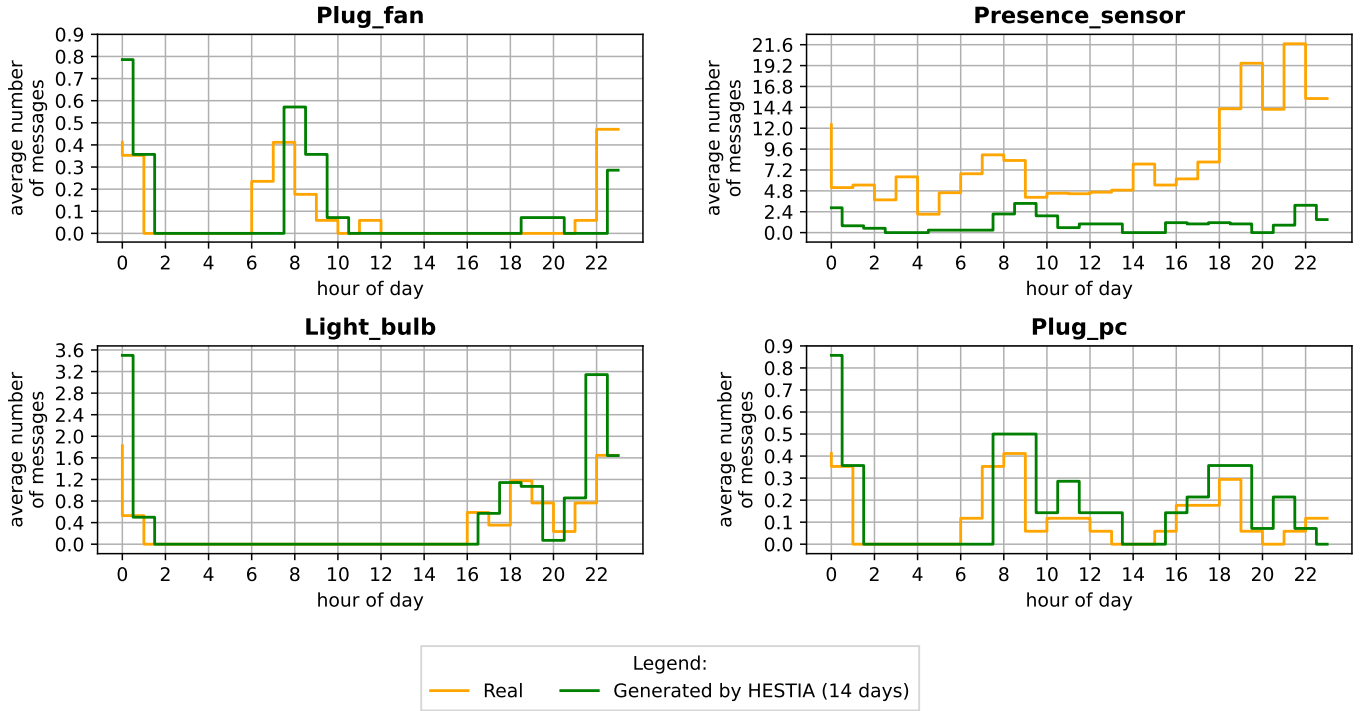


Fig. 9. Comparison of the average frequency of messages generated per device per hour between data from real devices over 14 days and data generated by HESTIA for 14 days.

TABLE VI

STATISTICAL ANALYSIS PERFORMED ON THE DATA FOR THE NUMBER OF MESSAGES GENERATED PER HOUR, WHERE. U : MANN-WHITNEY U TEST VALUE. $N_{synthetic}$ AND N_{real} : SAMPLE SIZES FOR THE TWO GROUPS. Z : STANDARDIZED STATISTIC VALUE, p : P VALUE ASSOCIATED WITH THE TEST INDICATES SIGNIFICANCE. GMR : AVERAGE RANKING FOR THE FIRST GROUP. AMR : AVERAGE RANKING FOR THE SECOND GROUP. INTERPRETATION: INDICATES WHETHER THE RESULT OF THE HYPOTHESIS TEST IS BASED ON THE P-VALUE.

Device	U	$N_{synthetic}$	N_{real}	Z	p	GMR	AMR
Plug_pc	58997.0	335	370	-1.794	0.073	361.89	344.95
Light_bulb	60843.5	335	370	-0.643	0.521	356.38	349.94
Presence_sensor	88238.5	335	370	10.642	0.000	274.60	423.98
Plug_fan	61197.5	335	370	-0.567	0.571	355.32	350.90

This analysis is interesting because it allows us to observe, for example, at which hours of the day the light bulb remains on for the longest time in the dataset collected from the real world, and later compare it with the data generated by the simulation performed by HESTIA. This comparison revealed a consistent behavior between the real and synthetic datasets, underscoring the simulator's effectiveness in replicating real-world patterns.

We also performed the Mann-Whitney U test to compare the distribution of device uptime between the 14-day generated data and the real data for the same period. However, now, the collection days were compared and analyzed, hour by hour, checking the number of minutes (0-60) each device remained in a "activation" state during these periods.

Table VII shows the results of the Mann-Whitney U test for each device analyzed. The results indicate that there is no statistically significant difference between the generated and real data for any of the devices. In all cases, the p-value was greater than 0.05, meaning that we cannot reject the null hypothesis that the frequency distributions of the generated

and real data are similar. Therefore, we conclude that the data generator successfully simulated the devices' usage patterns.

E. Analysis of temporal complexity

In this subsection, we provide a brief HESTIA complexity analysis in terms of time performance. We assume that HESTIA will simulate T units of time, the period of the whole simulation, I inhabitants, A activity per inhabitant, and the environment spaces graph $G(V, E)$ representing the rooms of a home has V vertices and E edges.

The HESTIA worst-case scenario occurs when each I inhabitant alternates between two or more different activities in two or more different spaces every constant step of time (e.i., x simulated second) throughout the whole simulation period (T units of time), requiring that HESTIA execute the search A^* algorithm $\frac{T}{x}$ times. It is well-known that A^* has worst-case performance $O(|E|\log|V|)$ time.

The time complexity of HESTIA is $O(T \cdot I \cdot A \cdot |E|\log|V|)$. HESTIA loops through all simulation time steps T , simulating

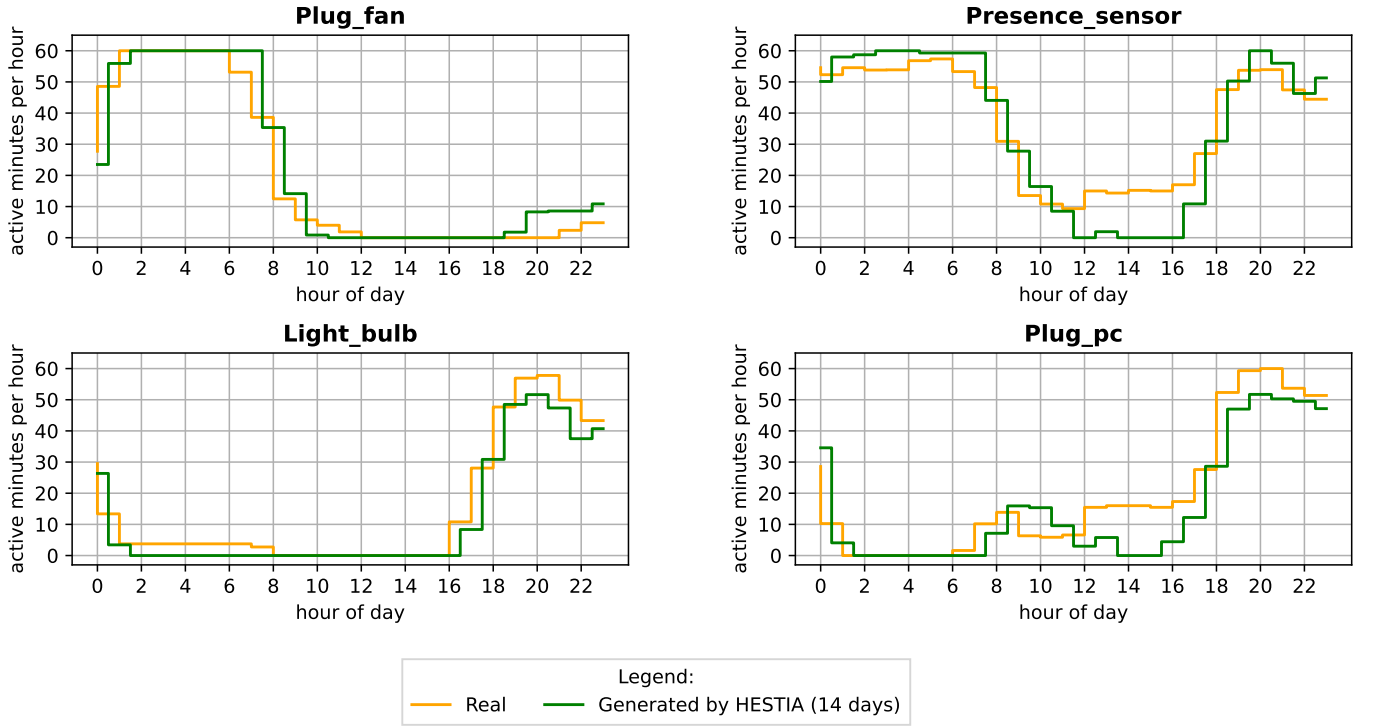


Fig. 10. Comparison of activation time of each device by time interval between data from real devices over 14 days and data generated by HESTIA for 14 days.

TABLE VII

STATISTICAL ANALYSIS PERFORMED ON THE DATA FOR ACTIVATION TIME PER HOUR, WHERE: U : MANN-WHITNEY U TEST VALUE. $N_{synthetic}$ AND N_{real} : SAMPLE SIZES FOR THE TWO GROUPS. Z : STANDARDIZED STATISTIC VALUE, p : P VALUE ASSOCIATED WITH THE TEST INDICATES SIGNIFICANCE. GMR : AVERAGE RANKING FOR THE FIRST GROUP. AMR : AVERAGE RANKING FOR THE SECOND GROUP. INTERPRETATION: INDICATES WHETHER THE RESULT OF THE HYPOTHESIS TEST IS BASED ON THE P-VALUE.

Device	U	$N_{synthetic}$	N_{real}	Z	p	GMR	AMR
Plug_pc	64668.5	336	371	1.005	0.315	347.03	360.31
Light_bulb	64809.5	336	371	1.124	0.261	346.61	360.69
Presence_sensor	58950.5	336	371	-1.283	0.199	364.05	344.90
Plug_fan	61177.0	336	371	-0.487	0.627	357.43	350.90

each I inhabitant activities and movements that multiply the costs A and $O(|E|\log|V|)$, respectively.

F. Discussion and limitations

The experimental evaluation of HESTIA revealed both its potential and limitations in modeling interactions in SHs. Through visual comparisons and statistical analyses, the simulator proved effective in reproducing general patterns of device behavior.

One of the key findings was the similarity between real and synthetic data generated by HESTIA, particularly in the temporal distribution of device interactions. Peaks in usage, such as presence detections in the morning and evening, were consistently replicated by the simulator. This alignment provides significant evidence of HESTIA's ability to capture real trends, which is crucial for any research related to SH and Internet of Things (IoT).

The statistical analysis revealed that the distributions of active time per hour showed significant similarity between

the real and synthetic data, with p-values greater than 0.05 for all four devices evaluated. In the comparison between the frequency of messages collected from the real environment and those generated by HESTIA, the similarity was confirmed in three of the four devices analyzed, which also had p-values greater than 0.05. This validates the use of HESTIA as a reliable tool for studies that do not require point-to-point correspondence but rather general behavioral representations.

However, while overall averages and general patterns were well reproduced, specific discrepancies arose, particularly related to the presence sensor. The presence sensor used to capture real data frequently produced false positives and negatives, complicating the analysis of this data compared to the data generated by HESTIA. The simulator is unable to replicate such high levels of noise without being specifically configured for it.

Although HESTIA stands out for its flexibility and effectiveness in simulating smart environments, some limitations require attention for future improvements. Firstly, the free-

dom provided to researchers to model routines can result in configurations that do not accurately reflect real-world device usage, especially if the defined routines are simplistic or incomplete, potentially compromising the representativeness of the simulated data. Furthermore, while the simulator allows users to include new product classes, a more extensive initial library of products would be a significant advancement.

Additionally, the environment configuration involves a large number of parameters, which can make the process increasingly complex as the sophistication of the simulated environment grows. The lack of a graphical interface further exacerbates this challenge, making it difficult to configure simulations, especially for users with less technical experience.

During the simulator's validation, only a limited number of devices were used, which restricted the generalization of the results to broader and more diverse scenarios. For this reason, HESTIA has been defined as a SH simulator rather than a general Smart Environments (SEs) simulator, as it was only validated in a residential environment.

VI. CONCLUSION

The research in the Smart Home (SH) field, such as the development and validation of machine learning models applied to this area, relies heavily on representative datasets. However, acquiring such datasets is challenging due to privacy and security concerns and the high costs and logistical difficulties associated with real-world data collection. Simulation of SHs presents a viable alternative, offering the ability to generate synthetic datasets that replicate real-world usage patterns while mitigating privacy risks and reducing costs.

This study introduces Home Environment Simulator Targeting Inhabitant Activities (HESTIA), a SHs simulator designed to tackle these challenges by generating data that accurately reflects interactions and behaviors observed in real-world settings. Built upon the message generation patterns of a major IoT device vendor, our simulator enables the creation of diverse and realistic routines, accommodating various device functionalities and inhabitant preferences. Additionally, it provides flexible output data from simulated devices, allowing researchers to simulate multiple scenarios and evaluate models according to their specific needs.

Besides being flexible, HESTIA is computationally efficient and statistically validated based on real-world data. With a time complexity of $O(T \cdot I \cdot A \cdot |E| \log |V|)$, HESTIA ensures efficient performance, even as the problem size grows significantly. Furthermore, the validation process includes the use of the Mann-Whitney U statistical hypothesis test.

Further enhancements could incorporate pre-defined scenes representing common environmental setups, broadening the generator's applicability and user-friendliness. Such improvements would facilitate more extensive and nuanced research, ultimately advancing the smart homes field and contributing to more effective, data-driven solutions for life activities, health, safety, and energy management. However, in its current form, HESTIA already offers a viable solution by enabling the generation of synthetic datasets that replicate real-world usage patterns, while also mitigating privacy risks and reducing research costs.

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